FLOW CHARACTERISTICS

Weir Accuracy

Laboratory conditions have shown thin-plate (http://openchannelflow.com/blog/weir-flow-terminology#thin-plate) weirs to be accurate to within +/-2%. However, unlike flumes, which are more forgiving to variances in upstream conditions and device condition, weirs are quite susceptible to these variations. Also, weirs require more maintenance (http://www.openchannelflow.com/blog/article/need-for-weir-maintenance) than flumes. Under field conditions, accuracies of +/- 5-15% are more common.

In particular, the crest (http://openchannelflow.com/blog/weir-flow-terminology#crest) of a weir must be kept in good condition – without nicks or abrasions and free of clinging debris.

The upstream weir pool (http://openchannelflow.com/blog/weir-flow-terminology#weir-pool) must also be of the proper size and kept free of sediments, debris, and trash that may change the depth of the weir pool.

Weir Flow Equations

V-notch Weirs [H = head (ft)]

WEIR ANGLE	CFS	GPM	MGD	L/S	M^3/HR
22.5°	0.497 $H^{2.5}$	223 $H^{2.5}$	0.321 $H^{2.5}$	274 $H^{2.5}$	988 $H^{2.5}$
30°	0.676 $H^{2.5}$	303 $H^{2.5}$	0.437 $H^{2.5}$	373 $H^{2.5}$	1,304 $H^{2.5}$
45°	1.04 $H^{2.5}$	465 $H^{2.5}$	0.669 $H^{2.5}$	571 $H^{2.5}$	2,060 $H^{2.5}$
60°	1.44 $H^{2.5}$	648 $H^{2.5}$	0.9333 $H^{2.5}$	797 $H^{2.5}$	2,870 $H^{2.5}$
90°	2.50 $H^{2.5}$	1,120 $H^{2.5}$	1.62 $H^{2.5}$	1,308 $H^{2.5}$	4,970 $H^{2.5}$
120°	4.33 $H^{2.5}$	1,904 $H^{2.5}$	2.80 $H^{2.5}$	2,390 $H^{2.5}$	8,610 $H^{2.5}$

Rectangular Weirs with End Contractions [L = length (ft), H = head (ft)]

cfs	3.33 (L-0.2H) $H^{1.5}$
gpm	1,500 (L-0.2H) $H^{1.5}$
mgd	2.15 (L-0.2H) $H^{1.5}$
l/s	1,840 (L-0.2H) $H^{1.5}$
m3/hr	6,620 (L-0.2H) $H^{1.5}$

Rectangular Weirs Without End Contractions [L = length (ft), H = head (ft)]

cfs	$3.33LH^{1.5}$

gpm	1,500 $LH^{1.5}$
mgd	2.15 $LH^{1.5}$
l/s	1,840 $LH^{1.5}$
m3/hr	6,620 $LH^{1.5}$

Cipolletti [L = length (ft), H = head (ft)]

cfs	$3.37LH^{1.5}$
gpm	$1,510LH^{1.5}$
mgd	2.18 LH

Weir Minimum / Maximum Head

V-notch Weirs

WEIR ANGLE	MINIMUM HEAD	MAXIMUM HEAD
22.5° - 120°	0.2' [6.096 cm]	2.0' [60.96 cm]

Rectangular Weir with End Contractions

CREST LENGTH	MINIMUM HEAD	MAXIMUM HEAD
1'	0.2' [6.096 cm]	0.5' [15.24 cm]
1.5'	0.2' [6.096 cm]	0.75' [22.86 cm]
2'	0.2' [6.096 cm]	1' [30.48 cm]
2.5'	0.2' [6.096 cm]	1.25' [38.10 cm]
3'	0.2' [6.096 cm]	1.5' [45.72 cm]
4'	0.2' [6.096 cm]	2' [60.96 cm]
5'	0.2' [6.096 cm]	2.5' [76.20 cm]
6'	0.2' [6.096 cm]	3' [91.44 cm]

8’	0.2’ [6.096 cm]	4’ [121.9 cm]
10’	0.2’ [6.096 cm]	5’ [152.4 cm]

Rectangular Weir Without End Contractions

CREST LENGTH	MINIMUM HEAD	MAXIMUM HEAD
1’	0.2’ [6.096 cm]	0.5’ [15.24 cm]
1.5’	0.2’ [6.096 cm]	0.75’ [22.86 cm]
2’	0.2’ [6.096 cm]	1’ [30.48 cm]
2.5’	0.2’ [6.096 cm]	1.25’ [38.10 cm]
3’	0.2’ [6.096 cm]	1.5’ [45.72 cm]
4’	0.2’ [6.096 cm]	2’ [60.96 cm]
5’	0.2’ [6.096 cm]	2.5’ [76.20 cm]
6’	0.2’ [6.096 cm]	3’ [91.44 cm]
8’	0.2’ [6.096 cm]	4’ [121.9 cm]
10’	0.2’ [6.096 cm]	5’ [152.4 cm]

Cipolletti

CREST LENGTH	MINIMUM HEAD	MAXIMUM HEAD
1’	0.2’ [6.096 cm]	0.5’ [15.24 cm]
1.5’	0.2’ [6.096 cm]	0.75’ [22.86 cm]
2’	0.2’ [6.096 cm]	1’ [30.48 cm]
2.5’	0.2’ [6.096 cm]	1.25’ [38.10 cm]
3’	0.2’ [6.096 cm]	1.5’ [45.72 cm]
4’	0.2’ [6.096 cm]	2’ [60.96 cm]
5’	0.2’ [6.096 cm]	2.5’ [76.20 cm]

6’	0.2’ [6.096 cm]	3’ [91.44 cm]
8’	0.2’ [6.096 cm]	4’ [121.9 cm]
10’	0.2’ [6.096 cm]	5’ [152.4 cm]

Weir Flow Tables

General free-flow discharge tables for thin-plate Weirs can be found in the Discharge Tables (https://www.openchannelflow.com/weirs/weir-plates/flow-characteristics) section.

(/) > Weirs (https://www.openchannelflow.com/weirs) > Fixed Weir Plates (https://www.openchannelflow.com/weirs/weir-plates) > Flow Characteristics



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